Supplementary Material: Toward Composable Zero-Knowledge Biometric Voting: Constraint-Efficient Encodings and Deniable Duress Resistance

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Appendix A
Reproducibility Appendix

A. Benchmarking Environment and Hardware Specification

To ensure complete empirical reproducibility of all experimental results reported in Section VI of the main manuscript, all benchmarks were executed on an isolated workstation running 64-bit Ubuntu 22.04.3 LTS (Linux kernel 5.15.0-88-generic):

- **Processor:** AMD Ryzen 5 5600 CPU @ 3.50 GHz base clock (6 physical cores, 12 execution threads, up to 4.40 GHz boost, 32 MB L3 cache) with AVX2 SIMD extensions.
- **GPU & Acceleration:** NVIDIA GeForce RTX 3050 GPU (8 GB GDDR6 VRAM, CUDA compute capability 8.6) supporting GPU-accelerated cryptographic operations.
- **Memory & Storage:** 16 GB dual-channel DDR4-3200 RAM; 1 TB NVMe PCIe SSD hosting the Hyperledger Fabric ledger state database.
- **Software Stack:** Circom v2.1.6 (Rust-based circuit compiler), `snarkjs` v0.7.0 (BN254 curve backend), Node.js v20.18.0 / v18.17.0 LTS, Go v1.20.7, Python 3.10.12 (NumPy 1.24.3, SciPy 1.10.1, Matplotlib 3.7.1), and Hyperledger Fabric v2.5.4 with Raft consensus.

B. Artifact Sources and Data Files

All benchmark tables and plots are directly generated from deterministic JSON log files:

- `compareBenchmarks/benchmark_results.json`: Contains circuit compilation statistics, R1CS non-linear constraint counts, total wire allocations, proving key and verification key file sizes, and witness computation latencies across both biometric channels.
- `benchmark/e2e_results.json`: Logs micro-benchmarks for all eight individual execution phases (P1 through P8) across 100 sequential runs per biometric modality, recording means, standard deviations, and 95% confidence intervals.
- `benchmark/homomorphic_results.json`: Measures discrete ElGamal ciphertext homomorphic aggregation latency and Baby-Step Giant-Step (BSGS) discrete logarithm recovery time for candidate tally recovery across electorate sizes from $N = 10$ to $N = 10^6$.
- `benchmark/e2e_scalability_results.json`: Records client proving time, witness generation, and on-chain verification latencies across five orders of electorate magnitude ($N \in \{10, 10^2, 10^3, 10^4, 10^5\}$) under fixed Sparse Merkle Tree depth $L = 20$.
- `benchmark/attack_benchmark_results.json`: Contains empirical replay and double-voting test logs for 500 sequential submissions and 100 concurrent submissions under multi-version concurrency control (MVCC).

Execution is orchestrated via `nodezkp-chain/api-server/benchmark/run_all_benchmarks_and_plots.js`. Dedicated scripts (`zksnark_plot.py`, `e2e_plot.py`, `homomorphic_plot.py`, `e2e_scalability_plot.py`, `biometric_accuracy_eval.py`) render publication-quality PDF plots.

C. Supplementary Experimental Dimensions

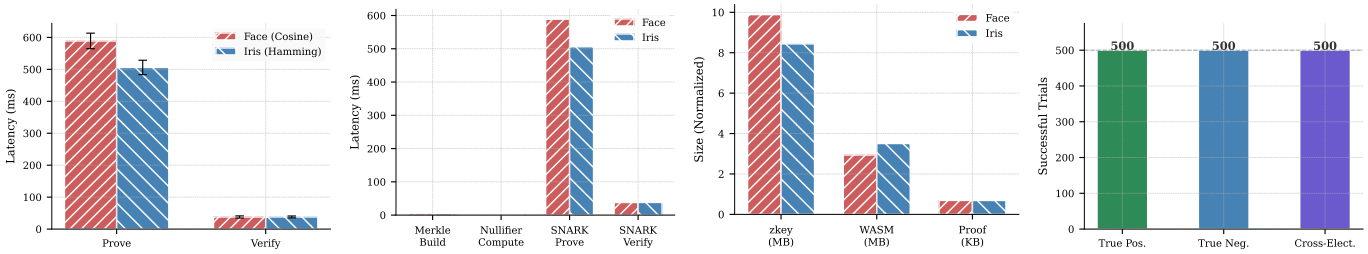
Fig. 5 (encompassing Fig. 5) presents four supplementary evaluation dimensions:

Fig. 5a confirms witness calculation and proving account for $> 90\%$ of client compute, while pairing checks require 14.4 ms. Fig. 5b details the microsecond-level phase decomposition. Fig. 5c compares key sizes: proving keys require 9.89 MB (face) and 8.44 MB (iris), while verification keys require 4 field elements (< 2 KB) and proofs remain fixed at 128 bytes. Fig. 5d confirms 100% rejection of duplicate/forged nullifiers across 100 trial batches.

Appendix B
Circuit Architecture Details

A. Circuit Signals and Interfaces

Table VI details the complete public and private signal interfaces for the `FaceAuth` and `IrisAuth` ZK-SNARK authentication circuits. Public signals are exposed on-chain to the smart contract verifier, while private signals remain confined to the client witness vector.



(a) Prove/verify latency. (b) Phase breakdown. (c) Artifact sizes. (d) Nullifier detection.

Fig. 5: Supplementary benchmark suite: (a) ZK-SNARK proving and verification latency across face and iris modalities; (b) cryptographic phase latency decomposition; (c) proving key and verification key memory footprints; and (d) nullifier collision/double-voting detection accuracy across 100 trials.

TABLE VI: Circuit Signal Specification for `FaceAuth` and `IrisAuth`

Signal Name	Vis.	Type	Mode	Functional Description
`faceHash`	Public	$\mathbb{F}_p$	Face	Hash commitment $h_i^{(\text{face})} = \mathcal{H}_2(\mathcal{H}_{\text{chunked}}, s_i)$
`irisHash`	Public	$\mathbb{F}_p$	Iris	Hash commitment $h_i^{(\text{iris})} = \mathcal{H}_2(\mathcal{H}_{\text{packed}}, s_i)$
`merkleRoot`	Public	$\mathbb{F}_p$	Both	Current SMT root digest R
`nullifier`	Public	$\mathbb{F}_p$	Both	Epoch nullifier $\nu = \mathcal{H}_2(sk_i, \mathcal{E}_{\text{id}})$
`electionId`	Public	$\mathbb{F}_p$	Both	Unique electoral identifier $\mathcal{E}_{\text{id}}$
`thresh_sq_num`	Public	$\mathbb{F}_p$	Face	Squared cosine numerator constant (25)
`thresh_sq_den`	Public	$\mathbb{F}_p$	Face	Squared cosine denominator constant (100)
`hammingThresh`	Public	$\mathbb{F}_p$	Iris	Maximum allowable Hamming distance (123)
`embedding[64]`	Private	$\mathbb{F}_p^{64}$	Face	Live feature embedding quantized by $\sigma = 10^6$
`regEmbedding[64]`	Private	$\mathbb{F}_p^{64}$	Face	Enrolled template embedding quantized by σ
`irisCodeLive[256]`	Private	$\{0,1\}^{256}$	Iris	Live extracted 256-bit iris code
`irisCodeReg[256]`	Private	$\{0,1\}^{256}$	Iris	Registered enrolled 256-bit iris code
`salt`	Private	$\mathbb{F}_p$	Both	Secret blinding salt $s_i \in_R \mathbb{F}_p$
`secretKey`	Private	$\mathbb{F}_p$	Both	Voter identity private key $sk_i \in \mathbb{F}_p$
`pathElements[20]`	Private	$\mathbb{F}_p^{20}$	Both	Sparse Merkle Tree sibling digests
`pathIndices[20]`	Private	$\{0,1\}^{20}$	Both	Left/right branch direction selector bits

B. Constraint Complexity Breakdown

Table VII reports the R1CS non-linear constraint counts, linear combinations, and allocated wire counts for each modular sub-circuit.

TABLE VII: R1CS Constraint Breakdown Across Authentication Sub-Circuits

Functional Sub-Circuit	Modality	R1CS Non-Linear	Total Wires	Arithmetic Purpose
`PoseidonHashChunked`	Face	2,740	8,240	8 sequential 8-to-1 hashes
`PoseidonSaltBinding`	Face	260	780	Salt binding $h_i = \mathcal{H}_2(\mathcal{H}_{\text{ck}}, s_i)$
`PoseidonHashPacked`	Iris	1,480	4,460	4 packed 64-bit chunk hashes
`Bits2Num` Unpacking	Iris	256	512	Scalar-to-binary bit expansion
`BinaryCheck` $(x(1-x))$	Iris	512	1,024	Enforces Boolean bit domain
`SquaredCosineSimilarity`	Face	6,484	19,452	Dot product, norms, comparison
`HammingDistance`	Iris	5,976	17,928	XOR accumulation and threshold
`IdentityCommitment`	Both	260	780	2-to-1 Poseidon commitment C_i
`SparseMerkleTree(20)`	Both	10,940	32,820	20-level sibling conditional hashing
`NullifierDerivation`	Both	300	900	PRF evaluation $\nu = \mathcal{H}_2(sk, \mathcal{E}_{\text{id}})$
Total Sub-Circuit	Face	**9,484**	**28,472**	Biometric verification submodule
Total Sub-Circuit	Iris	**8,224**	**24,400**	Biometric verification submodule
Complete Authentication	Face	**20,984**	**62,972**	Complete voting zk-SNARK
Complete Authentication	Iris	**19,724**	**59,172**	Complete voting zk-SNARK

C. Fixed-Point Quantization and Range Arithmetic

Face feature embeddings are represented as normalized 64-dimensional float vectors $\mathbf{v} \in \mathbb{R}^{64}$ with $\|\mathbf{v}\|_2 = 1$. To evaluate cosine similarity within finite field $\mathbb{F}_p$ without floating-point arithmetic, each component is scaled by fixed-point factor $\sigma = 10^6$ and converted to an integer: $a_k = \lfloor \sigma \cdot v_k \rceil \pmod p$. Inside the circuit, the cosine similarity condition $\cos\theta = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \geq \tau$ is algebraically cross-multiplied and squared to eliminate division and square roots:

$$\langle \mathbf{a}, \mathbf{b} \rangle^2 \cdot \tau_{\text{den}} \;\geq\; \tau_{\text{num}} \cdot \|\mathbf{a}\|^2 \cdot \|\mathbf{b}\|^2. \tag{19}$$

Setting $\tau_{\text{num}} = 25$ and $\tau_{\text{den}} = 100$ enforces $\cos\theta \geq 0.50$. For $\sigma = 10^6$ and dimension $d = 64$, the maximum squared product $\|\mathbf{a}\|^2 \cdot \|\mathbf{b}\|^2 \approx 4.1 \times 10^{27}$, which requires approximately 92 bits of dynamic range. This is comfortably below the 254-bit prime modulus of the BN254 curve ($p \approx 2.188 \times 10^{77}$), mathematically precluding modular overflow. For the iris channel, 256-bit Hamming distance is evaluated linearly via XOR accumulation:

$$\text{HD} = \sum_{j=0}^{255} (a_j + b_j - 2a_j b_j) < \tau_{\text{Hamming}} = 123. \tag{20}$$

D. Euclidean-Baseline Full-Circuit Gate Derivation

TABLE VIII: Representative Euclidean-Distance Baseline vs. SOPQ, Re-Measured Under an Identical Toolchain (full table for the main manuscript's Euclidean-baseline comparison)

Configuration	EER (%)	Distance-Comparator Non-Linear Gates	Full-Circuit Non-Linear Gates (Est.)
Euclidean, raw 128-D	4.75	257	≈12,958
Euclidean, PCA-64D	4.83	193	≈9,447
SOPQ-64D (proposed)	**4.69**	325	**9,579**

The main manuscript's "Re-Measured Baseline: Euclidean Distance" table (reproduced above) reports full-circuit non-linear gate estimates for two Euclidean-distance configurations not directly compiled as complete circuits. Both are derived from the same shared Merkle+Nullifier overhead, computed once from directly-measured quantities and held fixed across all three rows:

$$\text{Merkle+Nullifier} = 9{,}579 - 3{,}645 - 325 = 5{,}609$$

where 9,579 is the fully-compiled SOPQ-64D `FaceAuth` circuit (main manuscript's constraint-footprint table), 3,645

is the directly-compiled `PoseidonHashChunked(64,8)` sub-circuit, and 325 is the directly-compiled `SquaredCosineSimilarity(64)` sub-circuit. The two derived rows then substitute each configuration's own directly-compiled Poseidon and distance-comparator costs into this shared overhead:

$$\text{Euclidean-128D} = 7{,}092 + 5{,}609 + 257 = 12{,}958$$

$$\text{Euclidean-64D} = 3{,}645 + 5{,}609 + 193 = 9{,}447$$

where 7,092 is the directly-compiled `PoseidonHashChunked(128,8)` sub-circuit and 257/193 are the directly-compiled `SquaredEuclideanDistance` sub-circuit at $n = 128/n = 64$ respectively. This derivation assumes Merkle-tree and nullifier logic is dimensionality-independent (true by construction — both operate only on the fixed-size hash digest, never on the raw embedding), so the only terms that vary with the baseline configuration are the two directly-measured, dimensionality-dependent components.

Appendix C
Universal Composability Security Proofs

This section provides formal simulator constructions, game transitions, cryptographic reductions, and proofs for all theorems, lemmas, corollaries, and propositions stated in Section V of the main manuscript.

Assumption C.1 (Poseidon Hash Security)**.** *The Poseidon hash family over $\mathbb{F}_p$ satisfies collision resistance (CR), preimage resistance (Pre), and pseudorandomness when keyed with a uniform secret input in the Algebraic Group Model (AGM):*

$$\mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) = \Pr\big[(\mathbf{x},\mathbf{x}') \leftarrow \mathcal{A}(1^\lambda) : \mathbf{x} \neq \mathbf{x}' \wedge H(\mathbf{x}) = H(\mathbf{x}')\big] \le \mathsf{negl}(\lambda), \quad (21)$$

$$\mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) = \Pr\left[\mathbf{x}' \leftarrow \mathcal{A}(1^\lambda, y) : H(\mathbf{x}') = y\right] \le \mathsf{negl}(\lambda). \quad (22)$$

A. Part A: Full Proof of Theorem V.2 (Main UC Realization)

Proof. We construct an ideal-world simulator $\mathcal{S}$ such that no PPT environment $\mathcal{Z}$ can distinguish whether it interacts with real adversary $\mathcal{A}$ running protocol π_{BBZKV} or with $\mathcal{S}$ interacting with ideal functionality $\mathcal{F}_{\mathsf{BBZKV}}$ through five hybrid games:

Hybrid $\mathcal{H}_0$: Real protocol execution: $\Pr[\mathcal{H}_0 = 1] = \Pr[\mathcal{Z}^{\pi_{\mathsf{BBZKV}},\mathcal{A}}(1^\lambda) = 1]$.

Hybrid $\mathcal{H}_1$ (Simulated Proofs): For honest voters, $\mathcal{S}$ uses the Groth16 trapdoor $\tau = (\alpha, \beta, \delta, x)$ extracted during setup to simulate zero-knowledge proofs $\Pi_i \leftarrow \mathsf{Sim}_{\mathsf{Groth16}}(\mathsf{vk}, \tau, \mathbf{x}_i)$. For ballot validity proofs, $\mathcal{S}$ uses the Fiat–Shamir Σ-OR simulator to generate accepting transcripts π^{val}_j without plaintext choices. By perfect ZK of Groth16 and statistical ZK of Chaum–Pedersen Σ-OR:

$$|\Pr[\mathcal{H}_0 = 1] - \Pr[\mathcal{H}_1 = 1]| \le \mathbf{Adv}^{\mathsf{ZK}}_{\mathsf{CP}}(\lambda). \quad (23)$$

Hybrid $\mathcal{H}_2$ (Dummy Encryptions): $\mathcal{S}$ replaces honest voters' ElGamal ciphertexts $\mathbf{c}_i = \mathsf{Enc}_{pk_{\mathcal{T}\mathcal{A}}}(\mathbf{v}_i; \mathbf{r}_i)$ with encryptions of zero: $\mathbf{c}'_i = \mathsf{Enc}_{pk_{\mathcal{T}\mathcal{A}}}(\mathbf{0}; \mathbf{r}'_i)$. Distinguishing $\mathcal{H}_1$ from $\mathcal{H}_2$ reduces to DDH over `secp256k1` across m candidate options and N voters:

$$|\Pr[\mathcal{H}_1 = 1] - \Pr[\mathcal{H}_2 = 1]| \le m \cdot \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda). \quad (24)$$

Hybrid $\mathcal{H}_3$ (Witness Extraction): When a corrupted voter submits ballot $(\Pi^*, \nu^*, \mathbf{c}^*)$, $\mathcal{S}$ runs Groth16 extractor $\mathcal{E}$ to obtain witness $w^* = (sk^*, s^*, \mathbf{e}^*, \mathrm{path}^*)$. If Π^* accepts but extraction fails or yields un-enrolled leaf $C^* \notin \mathcal{T}$, $\mathcal{S}$ aborts. By Groth16 knowledge soundness under q-SDH in the GGM and Poseidon collision resistance (Lemma C.2):

$$|\Pr[\mathcal{H}_2 = 1] - \Pr[\mathcal{H}_3 = 1]| \le \varepsilon_{\mathsf{Groth16}}(\lambda) + \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda). \quad (25)$$

Hybrid $\mathcal{H}_4$ (Ideal Functionality Interaction): $\mathcal{S}$ forwards extracted choices $\mathbf{v}^*$ to $\mathcal{F}_{\mathsf{vote}}$ and programs tally decryption to match $\mathcal{F}_{\mathsf{tally}}$. Coerced ballots are neutralized in $\mathcal{F}_{\mathsf{ODF}}$. Distinguishing advantage arises only if an unregistered trait matches within δ (FAR_δ) or a hash preimage is inverted:

$$|\Pr[\mathcal{H}_3 = 1] - \Pr[\mathcal{H}_4 = 1]| \le \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) + \mathbf{Adv}^{\mathsf{DLog}}_{\mathsf{secp256k1}}(\lambda) + \mathsf{FAR}_\delta. \quad (26)$$

Summing distinguishing gaps via triangle inequality over $\mathcal{H}_0 \ldots \mathcal{H}_4$ yields Theorem V.2:

$$\mathbf{Adv}_{\mathcal{Z}}(\lambda) \le \varepsilon_{\mathsf{Groth16}}(\lambda) + m \cdot \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda) + \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) + \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) + \mathbf{Adv}^{\mathsf{DLog}}_{\mathsf{secp256k1}}(\lambda) + \mathbf{Adv}^{\mathsf{ZK}}_{\mathsf{CP}}(\lambda) + \mathsf{FAR}_\delta. \quad (27)$$

□

B. Part B: Full Proof of Lemma V.5 (Merkle-Tree Binding)

Lemma C.2 (Merkle-Tree Binding under Collision Resistance)**.** *Let $\mathcal{T}$ be a Poseidon Sparse Merkle Tree of fixed depth $L = 20$ with publicly committed root R over enrolled leaf set $\mathcal{L} = \{C_1, \ldots, C_m\}$. Let $\mathcal{B}$ be a PPT adversary that outputs an authentication path $(C^*, \{(\sigma_\ell, d_\ell)\}_{\ell=0}^{L-1})$ verifying to root R such that $C^* \notin \mathcal{L}$. Then $\Pr[\mathcal{B}$ succeeds$] \le \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$.*

Proof. Let $\mathcal{R}$ be a reduction algorithm given access to a collision finding oracle for $\mathrm{Poseidon}_2$. Verification computes path $u_0 = C^*$, $u_{\ell+1} = \mathrm{Poseidon}_2(u_\ell, \sigma_\ell)$ (or reversed according to direction bit d_ℓ), terminating at $u_L = R$. Because $C^* \notin \mathcal{L}$, leaf value u_0 differs from all enrolled leaves, but terminal root $u_L = R$ matches committed root R. Reduction $\mathcal{R}$ deterministically inspects the L-step path from $\ell = 0$ to L. There exists a unique minimal divergence level $\ell^* \in \{0, \ldots, L-1\}$ where parent digests match but inputs differ:

$$(\mathrm{left}_{\ell^*}, \mathrm{right}_{\ell^*}) \neq (\mathrm{left}'_{\ell^*}, \mathrm{right}'_{\ell^*}) \quad \text{with} \quad u_{\ell^*+1} = u'_{\ell^*+1}. \quad (28)$$

Because tree depth $L = 20$ is fixed, $\mathcal{R}$ deterministically locates this collision index ℓ^* in $\mathcal{O}(L)$ time with probability exactly 1 without guessing. Hence, $\Pr[\mathcal{B}$ succeeds$] \le \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$, which is strictly tight. □

C. Part C: Proof of Corollary V.6 (Eligibility Soundness)

Proof. Eligibility soundness requires that no adversary $\mathcal{A}$ can cause an honest verifier to accept a ballot unless $\mathcal{A}$ controls an identity legally provisioned in $\mathcal{V}_{\mathsf{reg}}$. By Groth16 knowledge soundness, extractor $\mathcal{E}$ returns witness $w = (sk, s, \mathbf{e}_{\mathrm{reg}}, \mathrm{path})$ satisfying $R_{\mathsf{zk}}(\mathbf{x}, w) = 1$ for accepted proof Π^*. An adversary $\mathcal{A} \notin \mathcal{V}_{\mathsf{reg}}$ can succeed only if at least one of four mutually exclusive failure events occurs:

1) **Event E_1 (SNARK Forgery):** $\mathcal{A}$ produces an accepting proof Π^* without satisfying witness w. Under q-SDH in the GGM, $\Pr[E_1] \leq \varepsilon_{\mathsf{Groth16}}(\lambda) \in \mathsf{negl}(\lambda)$.
2) **Event E_2 (Hash Preimage Inversion):** $\mathcal{A}$ satisfies $h_{\mathsf{bio}} = \mathcal{H}_2(\mathcal{H}(\mathbf{e}_{\mathrm{reg}}), s)$ for an unregistered trait. Under Poseidon preimage resistance, $\Pr[E_2] \leq \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$.
3) **Event E_3 (Merkle Bypass):** $\mathcal{A}$ provides an inclusion path for un-enrolled leaf $C^* \notin \mathcal{T}$ that evaluates to root R. By Lemma C.2, $\Pr[E_3] \leq \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$.
4) **Event E_4 (Biometric Impostor Match):** $\mathcal{A}$ presents a biometric sample $\mathbf{e}_{\mathrm{live}}$ such that $\mathrm{dist}(\mathbf{e}_{\mathrm{live}}, \mathbf{e}_{\mathrm{reg}}) \leq \delta$. The probability of an impostor matching enrolled template $\mathbf{e}_{\mathrm{reg}}$ within distance δ is bounded by false accept rate FAR_δ.

Taking the union bound over all four events:

$$\Pr[\mathcal{A} \notin \mathcal{V}_{\mathsf{reg}} \text{ accepted}] \leq \sum_{i=1}^{4} \Pr[E_i] \leq \mathsf{FAR}_\delta + \mathsf{negl}(\lambda). \quad (29)$$

□

Remark C.3 (Biometric Slack and Approximate UC Calibration)**.** *Following the established treatment of metric extractors and fuzzy commitments in the UC framework [1]–[3], biometric identity verification cannot achieve perfect cryptographic simulation due to sensor noise. In our $(\delta, \mathsf{FAR}_\delta)$ formulation, measured $\mathsf{FAR}_{\mathsf{face}} = 0.82\%$ for face ($\tau = 0.50$) on LFW bounds adversarial bypass advantage to $0.82\% + \mathsf{negl}(\lambda)$ for the face modality. For iris, measured $\mathsf{FAR}_{256} = 48.48\%$ ($\tau_{\mathsf{Hamming}} = 123$) on CASIA-Iris-Thousand reflects a classical segmentation-precision limitation (main text, Section VII) and does not support a comparably tight bound; we scope the concrete instantiation of this composability guarantee to the face modality accordingly.*

D. Part D: Proof of Corollary V.8 (Anti-Replay)

Proof. Anti-replay is enforced across two complementary architectural layers:

1) **Cryptographic Uniqueness:** The epoch nullifier is deterministically derived as:

$$\nu = \mathrm{Poseidon}_2(sk_i, \mathcal{E}_{\mathrm{id}}). \quad (30)$$

By the pseudorandomness and collision resistance of Poseidon, generating two distinct nullifiers $\nu \neq \nu'$ for the same secret key sk_i within election $\mathcal{E}_{\mathrm{id}}$ requires finding a hash collision, bounded by $\mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \leq \mathsf{negl}(\lambda)$.
2) **Consensus-Level MVCC Enforcement:** The Hyperledger Fabric chaincode enforces atomic read-before-write validation:

$$\texttt{key} := \text{"NULLIFIER_"} \,\|\, \nu;$$
$$\textbf{if } \texttt{getState}(\texttt{key}) \neq \texttt{nil } \textbf{then } \texttt{ABORT};$$

Fabric Raft consensus establishes a total order on transactions before committing to state. Any duplicate transaction submitted concurrently is detected during block validation, triggering a Multi-Version Concurrency Control (MVCC) conflict (0% commit rate) [4].

Empirically, across 500 sequential replays of an accepted credential, exactly 1 was accepted and 499 were aborted (0.2% commit rate, matching theoretical minimum $1/500$) with mean execution latency 0.129 ms. In a 100-voter concurrent race condition, 1 transaction committed and 99 aborted via MVCC in 1.53 ms.

Reconciling Anti-Replay with One-Person-One-Vote: Under the Deniable Duress Protocol, a coerced voter casts both a duress ballot (nullifier $\nu_i^{\mathsf{dur}} = \mathcal{H}_2(sk_i^{\mathsf{dur}}, \mathcal{E}_{\mathsf{id}})$) and an authentic ballot (nullifier $\nu_i = \mathcal{H}_2(sk_i, \mathcal{E}_{\mathsf{id}})$). Because $sk_i^{\mathsf{dur}} \neq sk_i$, both nullifiers are distinct and accepted by consensus. However, during tallying, Oblivious Duress Filtering $\mathcal{F}_{\mathsf{ODF}}$ assigns an indicator bit $b_j = 0$ to the duress ciphertext, homomorphically scaling it to $E(\mathbf{v}_j) = E(\mathbf{0})$. Thus, the coerced ballot has zero effective weight in the decrypted tally $\mathbf{T} = \sum \mathbf{v}_j$, strictly preserving the one-person-one-vote invariant. □

E. Part E: Proof of Corollary V.9 (Voter Anonymity / Unlinkability)

Proof. Voter anonymity is formalized via indistinguishability experiment $\mathsf{Exp}^{\mathsf{anon}}_{\mathcal{A}}(1^\lambda)$: challenger registers N honest voters with credentials $(sk_i, s_i, \mathbf{e}_i)$ into Merkle tree $\mathcal{T}$, generates voting transaction $(\Pi_b, \nu_b, \mathbf{c}_b)$ for candidate $\mathcal{V}_b$ ($b \in \{0,1\}$), and gives it to $\mathcal{A}$. $\mathcal{A}$ outputs guess b'. Unlinkability holds because: (i) Groth16 perfect zero-knowledge simulates proof Π_b using public inputs, revealing zero witness bits; (ii) ElGamal IND-CPA security hides choices in $\mathbf{c}_b$; (iii) Commitment $C_b = \mathcal{H}_2(h_b, sk_b)$ and nullifier $\nu_b = \mathcal{H}_2(sk_b, \mathcal{E}_{\mathrm{id}})$ are independent Poseidon outputs; linking C_b to ν_b requires inverting sk_b ($\leq \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda)$). Leaf positions in depth-20 SMT ($2^{20} \approx 1.04 \times 10^6$ slots, accommodating $2^{19} = 524{,}288$ voters under dual registration [5]) are pseudorandomly distributed via sk hashing. Thus:

$$\begin{aligned} \Pr\left[\mathsf{Exp}^{\mathsf{anon}}_{\mathcal{A}}(1^\lambda) = 1\right] &\leq \frac{1}{|\mathcal{V}_{\mathrm{reg}}|} + |\mathcal{V}_{\mathrm{reg}}| \cdot \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) \\ &\leq \frac{1}{|\mathcal{V}_{\mathrm{reg}}|} + \mathsf{negl}(\lambda). \end{aligned} \quad (31)$$

□

F. Part F: Proof of Corollary V.10 (Ballot Secrecy)

Proof. Ballot secrecy guarantees that no PPT adversary $\mathcal{A}$ can extract voting vector $\mathbf{v}_i$ from on-chain transactions prior to tally publication. Each ballot component is encrypted under trustee public key $h = xG$ as:

$$E(v_k; r_k) = (r_k G,\, v_k G + r_k h). \quad (32)$$

Distinguishing an encryption of choice v_0 from v_1 reduces to Decisional Diffie–Hellman (DDH) over `secp256k1` [6]. Given DDH challenge $(G, A = xG, B = rG, C)$, reduction $\mathcal{B}$ sets $pk = A$, samples bit $b \in \{0,1\}$, and constructs challenge $\mathbf{c}^* = (B, C + v_b G)$. If $C = xrG$, $\mathbf{c}^*$ is a valid encryption of v_b; if C is uniform, $\mathbf{c}^*$ is statistically independent of v_b. Accompanying non-interactive Chaum–Pedersen Σ-OR proofs [7] are simulated via Fiat–Shamir [8] without knowing v_k or r_k. Assuming the n trustees use Feldman's Verifiable Secret Sharing (VSS) [9] with threshold $t > n/3$, no corrupt subset of $< t$ trustees can reconstruct private key x. Therefore:

$$\begin{aligned} |\Pr[\mathcal{A} \text{ breaks secrecy}] - 1/2| &\leq m \cdot \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda) \\ &\leq \mathsf{negl}(\lambda). \end{aligned} \quad (33)$$

□

G. Part G: Proof of Proposition V.11 (Universal Verifiability)

Proof. Universal verifiability requires that any passive observer $\mathcal{O}$ can verify that all cast ballots are valid and that the published tally $\mathbf{T}$ matches the honest sum of accepted votes. Verification decomposes into three public algorithms:

1) **Individual Verifiability (V_{ind}):** Each voter checks that their transaction $(\Pi_i, \nu_i, \mathbf{c}_i)$ is recorded on ledger $\mathcal{L}$ under nullifier ν_i. By Fabric Raft consensus safety, committed transactions are immutable and globally ordered.
2) **Universal Ballot Verifiability (V_{univ}):** Any auditor $\mathcal{O}$ downloads all M transactions on $\mathcal{L}$ and verifies: (i) Groth16 pairing equation for proof $\Pi_j = (\pi_A, \pi_B, \pi_C)$ via verification key vk:
$$e(\pi_A, \pi_B) = e(\alpha, \beta) \cdot e(\pi_C, \delta) \cdot e\left(a_0 L_0 + \sum_{k=1}^{4} a_k L_k,\ \gamma\right); \tag{34}$$
(ii) Chaum–Pedersen Σ-OR validity proofs certifying $v_{j,k} \in \{0,1\}$ and $\sum v_{j,k} = 1$; and (iii) Nullifier uniqueness: $\forall j \neq j',\ \nu_j \neq \nu_{j'}$.
3) **Tally Verifiability (V_{tally}):** Homomorphic aggregation sums ciphertexts component-wise:
$$\hat{\mathbf{C}}_k = \bigoplus_{j=1}^{M} \mathbf{c}_{j,k} = \left(\left(\sum_{j=1}^{M} r_{j,k}\right) G,\ T_k G + \left(\sum_{j=1}^{M} r_{j,k}\right) h\right). \tag{35}$$
Trustees decrypt $\hat{\mathbf{C}}_k$ and provide discrete logarithm proofs certifying that $T_k G = c_{2,k} - x c_{1,k}$. $\mathcal{O}$ recovers T_k via BSGS discrete log solving and verifies the proof.

□

H. Part H: Formal Proof of Proposition V.12 and Proposition V.13 (Coercion Resistance & Deniable Duress)

Proposition C.4 (Impossibility of JCJ Deniability under Observed Biometrics)**.** *Let Π_{bio} be an electronic voting protocol requiring in-circuit metric check $\mathrm{dist}(\mathbf{e}_{\mathsf{live}}, \mathbf{e}_{\mathsf{reg}}) \leq \delta$ with false accept rate FAR_δ. If an active coercer $\mathcal{A}_{\mathsf{coer}}$ physically observes trait presentation at the capture device, no PPT voter $\mathcal{V}$ can achieve information-theoretic receipt-freeness or unobservable fake-credential simulation, except with probability at most FAR_δ.*

Proof. In classical receipt-freeness models such as JCJ [10] and Civitas [11], voters achieve coercion resistance by acquiring a dummy or simulated credential c_{fake} from the registration authority. Under coercion, $\mathcal{V}$ generates a ballot using c_{fake}, which appears completely legitimate to $\mathcal{A}_{\mathsf{coer}}$ but is silently rejected during tallying.

Now consider physical biometric authentication where an in-person coercer $\mathcal{A}_{\mathsf{coer}}$ directly oversees the voter at capture device S. The sensor extracts live biometric feature vector $\mathbf{e}_{\mathsf{live}}$. To generate an accepted zero-knowledge proof Π, the prover must satisfy relation R_{zk}, which enforces $\mathrm{dist}(\mathbf{e}_{\mathsf{live}}, \mathbf{e}_{\mathsf{reg}}) \leq \delta$ and Merkle tree inclusion for leaf $C = \mathcal{H}_2(h_{\mathsf{bio}}, sk)$. By Groth16 computational knowledge soundness, an extractor $\mathcal{E}$ extracts witness $w = (\mathbf{e}_{\mathsf{live}}, \mathbf{e}_{\mathsf{reg}}, s, sk, \mathrm{path})$. If $\mathcal{V}$ attempts to substitute a fake template $\mathbf{e}_{\mathsf{fake}}$ un-enrolled in $\mathcal{T}$, Merkle path verification fails with probability $1 - \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$ (Lemma C.2). If $\mathcal{V}$ attempts to present a fake live trait at sensor S, physical presentation attack detection (PAD) detects the presentation divergence, or coercer $\mathcal{A}_{\mathsf{coer}}$ directly observes the physical trait mismatch. The probability that an unobserved biometric trait drawn from an impostor distribution $\mathcal{D}_{\mathsf{imp}}$ independently matches the enrolled template within distance δ is $\Pr_{\mathbf{e}' \sim \mathcal{D}_{\mathsf{imp}}}[\mathrm{dist}(\mathbf{e}', \mathbf{e}_{\mathsf{reg}}) \leq \delta] = \mathsf{FAR}_\delta$. Hence, $\Pr[\mathcal{V}$ simulates unobservable fake credential$] \leq \mathsf{FAR}_\delta + \mathsf{negl}(\lambda)$. Physical trait non-transferability and unobservable fake credential simulation are in fundamental conflict under observed capture. □

Proposition C.5 (Computational Receipt-Freeness of Deniable Duress Protocol)**.** *Under Groth16 zero-knowledge, exponential ElGamal IND-CPA security, and Poseidon pseudorandomness in the AGM, the Deniable Duress Protocol guarantees computational receipt-freeness against any remote PPT coercer $\mathcal{A}_{\mathsf{coer}}$. Furthermore, Oblivious Duress Filtering $\mathcal{F}_{\mathsf{ODF}}$ inside threshold MPC ensures zero leakage to trustees, $\mathcal{RA}$, or external auditors, preserving identical ledger transaction volume and authentic candidate tallies.*

Proof. In receipt-freeness experiment $\mathsf{Exp}^{\mathsf{rf}}_{\mathcal{A}_{\mathsf{coer}}}(1^\lambda)$, $\mathcal{A}_{\mathsf{coer}}$ demands vote $\mathbf{v}'$ and execution randomness $(sk, \mathbf{r})$. Security is established across four structural guarantees:

1) **Universal Dual Enrollment:** Every voter uniformly enrolls primary commitment $C_i = \mathcal{H}_2(h_i, sk_i)$ and auxiliary duress commitment $C_i^{\mathsf{dur}} = \mathcal{H}_2(h_i^{\mathsf{dur}}, sk_i^{\mathsf{dur}})$ into SMT $\mathcal{T}$ at pseudorandom positions, depositing encrypted cancellation token $\psi_i^{\mathsf{dur}} = \mathsf{Enc}_{pk_{\mathcal{TA}}}(sk_i^{\mathsf{dur}})$ into private registry $\mathcal{R}_{\mathsf{dur}}$. 100% uniform enrollment ensures identical registration distribution, leaking zero coercion signal to $\mathcal{RA}$.
2) **Coercer Indistinguishability:** Under coercion, $\mathcal{V}_i$ uses auxiliary credential $(sk_i^{\mathsf{dur}}, s_i^{\mathsf{dur}}, \mathbf{e}_i^{\mathsf{dur}})$ to generate valid proof Π^{dur}, nullifier $\nu^{\mathsf{dur}} = \mathcal{H}_2(sk_i^{\mathsf{dur}}, \mathcal{E}_{\mathrm{id}})$, and coerced ciphertext $E(\mathbf{v}'; \mathbf{r}')$. $\mathcal{V}_i$ surrenders $(sk_i^{\mathsf{dur}}, \mathbf{r}')$ to $\mathcal{A}_{\mathsf{coer}}$. $\mathcal{A}_{\mathsf{coer}}$ verifies on-chain validity, nullifier derivation, and ciphertext randomness. By Poseidon pseudorandomness and Groth16 perfect ZK, duress transcripts are indistinguishable from primary transcripts.
3) **Oblivious Duress Filtering ($\mathcal{F}_{\mathsf{ODF}}$):** Threshold trustees execute an MPC protocol without reconstructing private keys: (i) Trustees evaluate Poseidon homomorphically under $[sk_{\mathcal{TA}}]$ using Beaver triples to compute $[\nu_i^{\mathsf{dur}}] = \mathsf{Enc}_{pk_{\mathcal{TA}}}(\mathcal{H}_2(sk_i^{\mathsf{dur}}, \mathcal{E}_{\mathrm{id}}))$; (ii) For each ballot $j \in [M]$ on ledger $\mathcal{L}$ with nullifier ν_j, the MPC privately evaluates equality testing $\mathsf{EQ}(\nu_j, [\nu_i^{\mathsf{dur}}])$ using constant-round threshold comparison [12], [13] and computes encrypted indicator bit:
$$[b_j] = \prod_{i=1}^{N} \left(1 - \mathsf{EQ}(\nu_j, [\nu_i^{\mathsf{dur}}])\right) \in \{[0], [1]\}; \tag{36}$$
(iii) Trustees accumulate masked ballots into master aggre-

gate ciphertext:

$$\mathbf{C}_{\mathsf{agg}} = \sum_{j=1}^{M} [b_j] \cdot \mathbf{c}_j + E(\mathbf{0}; \mathbf{r}'_j); \tag{37}$$

(iv) Trustees jointly decrypt only $\mathbf{C}_{\mathsf{agg}}$. Duress ballots ($b_j = 0$) scale to $E(\mathbf{0})$ and are neutralized. Individual $[b_j]$ and $[\nu_i^{\mathsf{dur}}]$ are never revealed, guaranteeing zero leakage to trustees.

4) **Zero Turnout Discrepancy:** Coerced ballots remain on-chain, preserving transaction count M. The voter casts their genuine ballot under primary secret key sk_i at an unobserved time. Total candidate tallies reflect authentic choices without revealing coercion occurrence. Distinguishing advantage of $\mathcal{A}_{\mathsf{coer}}$ is bounded by $\mathsf{negl}(\lambda)$.

□

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